

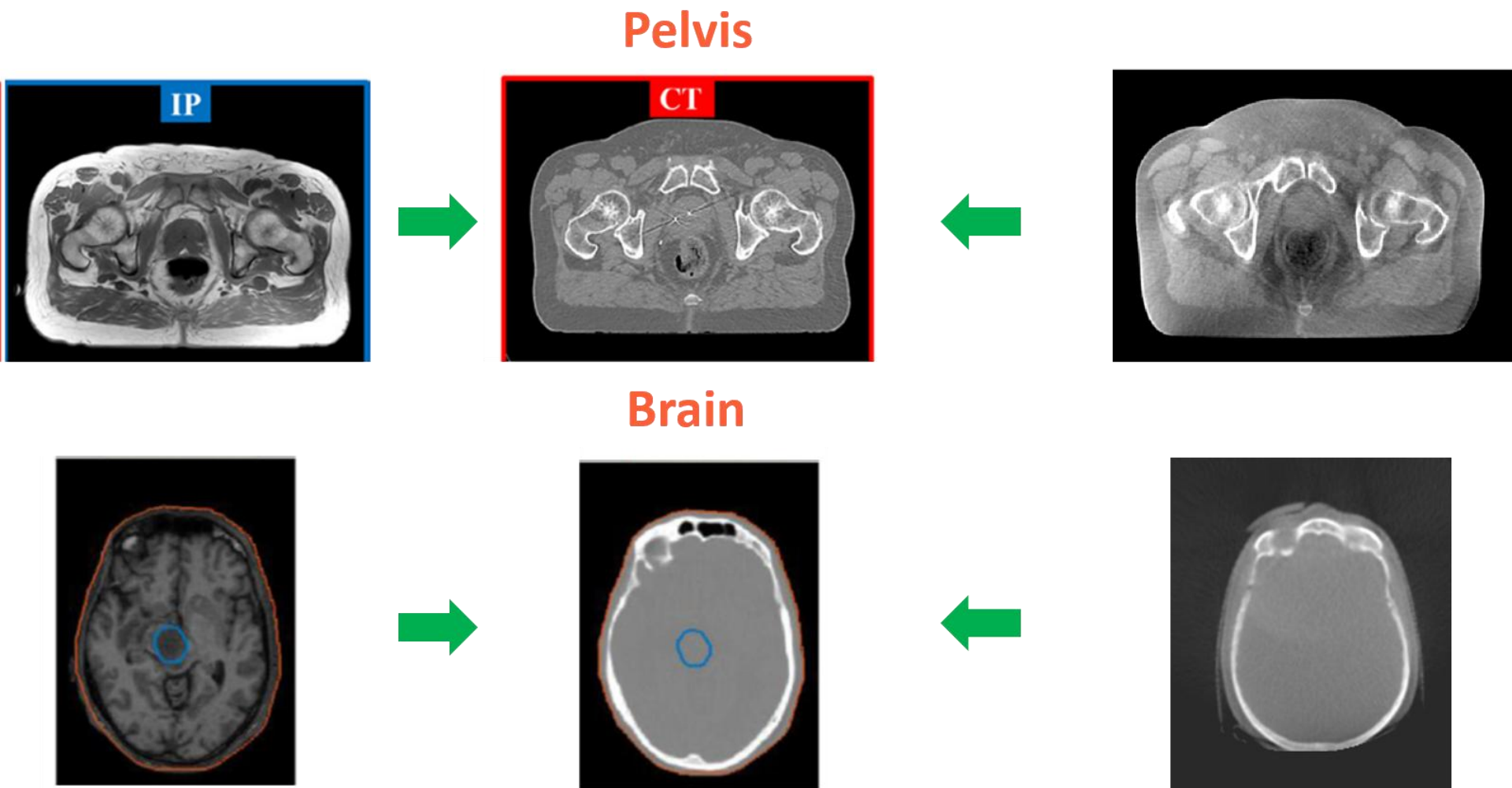
Test Task 2 - CBCT Leaderboard

#	User (Team)	Algorithm	Created	Final rank
1st	 whisneyzyw  (SMU-MedVision)	SynthRAD2023	27 July 2023	1.0000
2nd	 elliottqii  (GEneRaTion)	cbct2ct	22 Aug. 2023	2.0000

SynthRAD2023 Top 1 Method
Transfer learning and 2.5D UNet++ for CBCT-CTSynthesis in SynthRAD2023

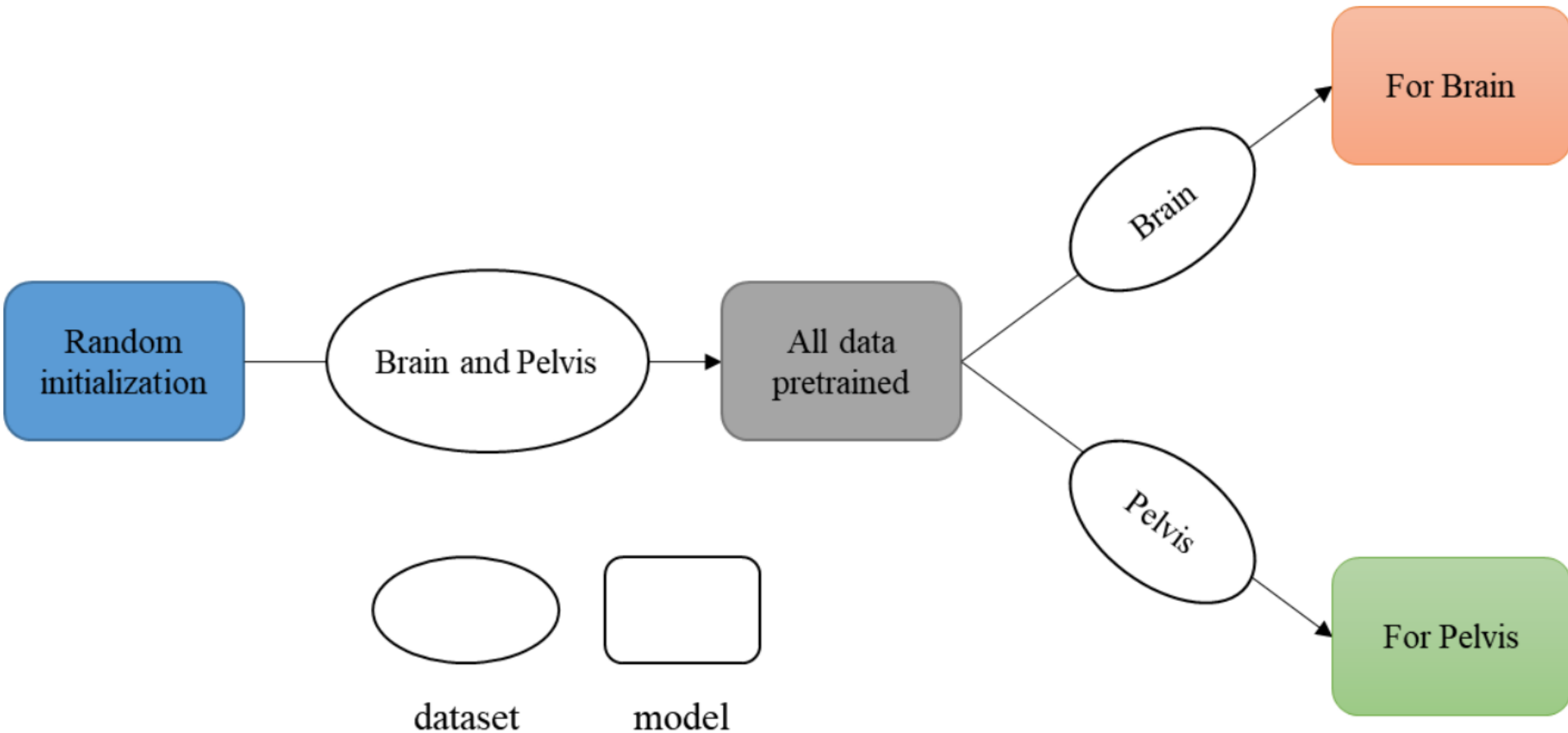
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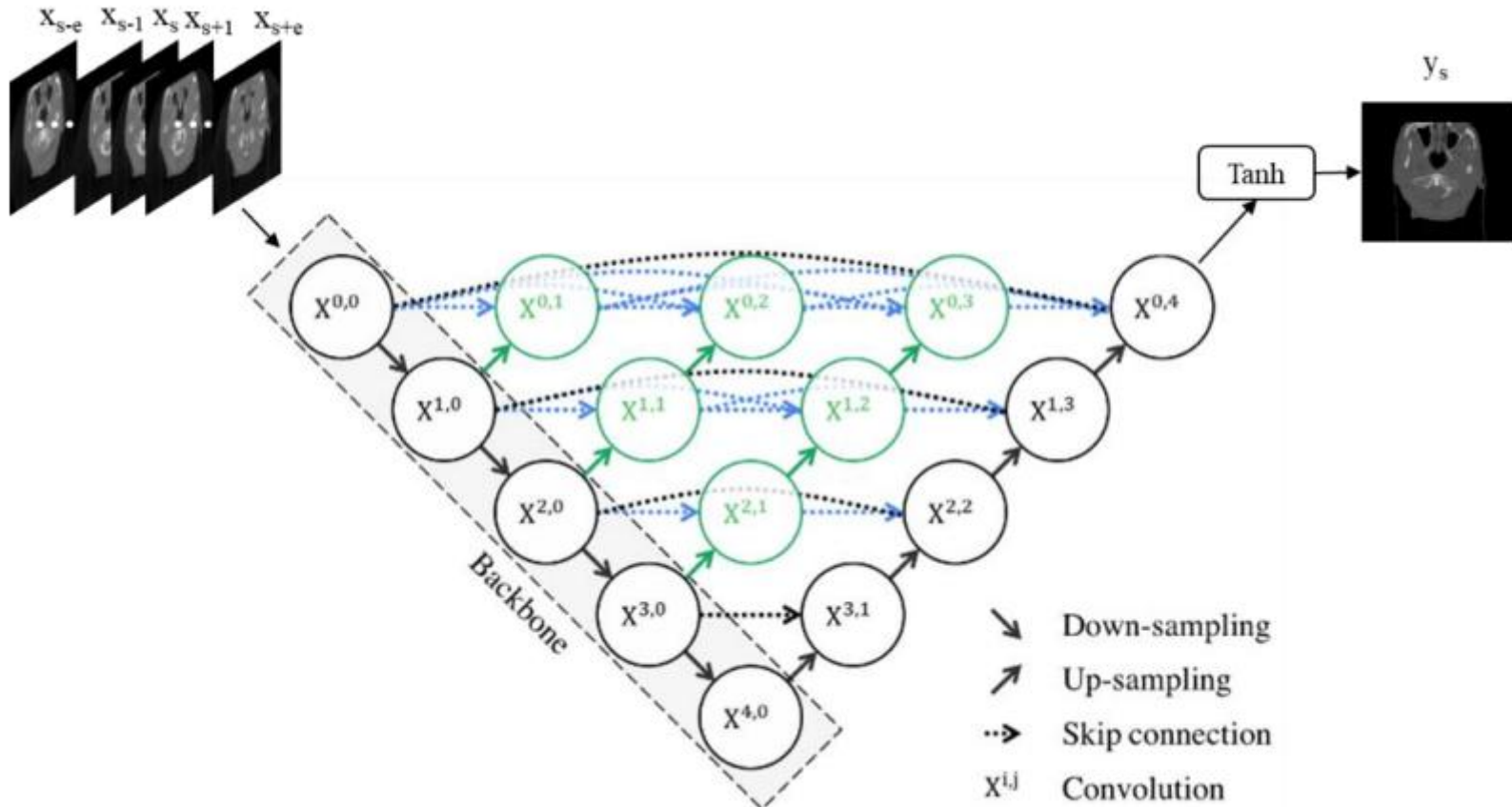
Methodology

Transfer learning



Methodology

Network architecture



Methodology

Loss function, Pre-Processing, Post-Processing

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{MAE}} + \lambda_2 \mathcal{L}_{\text{MSE}} + \lambda_3 \mathcal{L}_{\text{perc}} + \lambda_4 \mathcal{L}_{\text{SSIM}}$$

- Pre-Processing
 - Resize – maintaining original aspect ratio.
 - Clipping – CBCT: 0~99.5% / CT: 1000 HU ~ 3000 HU
 - Normalization – Linear Normalization (-1~1)
- Post-Processing
 - Test Time Augmentation(TTA) – Horizontal & Vertical flips
 - Model Ensembling - equal contribution & final result was the average of their predictions.

Experiments

Hyperparameter

- Data-split – 5 fold
- Transfer learning – True
- Network frameworks – U-Net++
- Extended slices – e=2
- Weights for loss functions – Unet++(ResNet101)
- Normalization – Linear Normalization
- Learning rate decay strategy – Step

Table 1. Experimental results on transfer learning (Brain-fold0).

Transfer learning	MAE↓	SSIM↑	PSNR↑
False	47.4116	0.9122	28.8336
True	46.7517	0.9132	28.9282

Experiments

Hyperparameter

Table 2. Experimental results on network frameworks (Brain-fold0).

Frameworks	Backbones	MAE↓	SSIM↑	PSNR↑
UNet	ResNet34	49.2108	0.9089	28.6349
UNet	ResNet50	49.1836	0.9094	28.6297
UNet	SE-ResNet50	49.3769	0.9089	28.6373
UNet	ResNet101	48.6226	0.9102	28.7447
DeepLab V3+	ResNet34	72.5901	0.8785	26.0368
MANet	ResNet50	48.7676	0.9100	28.7010
MANet	ResNet101	48.6942	0.9098	28.7868
UNet++	ResNet50	47.5455	0.9120	28.8422
UNet++	ResNext101	47.2059	0.9124	28.9149

2.4 Extended slices

Table 3. Experimental results on number of expanded slices (Brain-fold0).

Number of expanded slices	MAE↓	SSIM↑	PSNR↑
$e = 0$	50.5087	0.9074	28.4707
$e = 1$	49.4986	0.9087	28.6085
$e = 2$	49.2108	0.9089	28.6349
$e = 3$	49.2585	0.9088	28.6424
$e = 4$	49.4012	0.9086	28.6377

2.5 Weights for loss functions

Table 4. Experimental results on weights of loss functions (Brain-fold0).

Networks	λ_1	λ_2	λ_3	λ_4	MAE↓	SSIM↑	PSNR↑
UNet (ResNet34)	1	0	0	0	50.2554	0.9071	28.4748
UNet (ResNet34)	0	1	0	0	51.0785	0.9063	28.6762
UNet (ResNet34)	10	0	1	0	49.8559	0.9086	28.6305
UNet++ (ResNext101)	10	0	1	0	46.7517	0.9132	28.9282
UNet++ (ResNext101)	10	0	1	0.5	46.9727	0.9127	28.9006
UNet++ (ResNext101)	10	0	1	1	46.9738	0.9128	28.8908

Table 5. Experimental results on size normalization (Brain-fold0).

Size	MAE↓	SSIM↑	PSNR↑
(256, 256)	50.2554	0.9071	28.4748
(288, 288)	49.2108	0.9089	28.6349
(320, 320)	49.6512	0.9086	28.5688

Table 6. Experimental results on size normalization (Pelvis-fold0).

Size	MAE↓	SSIM↑	PSNR↑
(416, 352)	56.4080	0.7934	27.2876
(416, 384)	55.6471	0.7973	27.1237
(416, 416)	53.6223	0.8020	27.6472

Table 7. Experimental results on normalization modes (Brain-fold0).

Modes	MAE↓	SSIM↑	PSNR↑
Linear normalization	50.5936	0.9074	28.4536
Z-score	51.0680	0.9068	28.3819

2.7 Learning rate decay strategy

Table 8. Experimental results on learning rate decay strategy (Brain-fold0).

Strategies	MAE↓	SSIM↑	PSNR↑
Step	50.5087	0.9074	28.4707
ploy	50.5936	0.9074	28.4536

Conclusion

The final confirmed hyper-parameters for training the models

Hyper-parameters	Brain	Pelvis
Network	UNet++ (ResNext101)	
Normalization	Linear	
Number of expanded slices	2	
Data augmentation	Shift Scale Rotate+ Horizontal Flip+ Vertical Flip	
Loss function	$1\mathcal{L}_{MAE} + 20\mathcal{L}_{perc}$	
Learning rate decay strategy	step	
Initial learning rate	0.0002	
L2 regularization	0.0001	
Optimizer	AdamW	
TTA	True	
Batchsize	8	4
Training epoch	50	40
Input size	(288, 288)	(416, 416)